

Homework 3

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Problem 1

1.1 Part A

Given the modified version of Stoke's first problem we are asked to find the equation that governs the x-component of velocity for the fluid above the plate. The boundary conditions we are given are as follows

$$y = 0, \quad \mathbf{u} = [U, V]$$

$$y \rightarrow \infty, \quad \mathbf{u} \rightarrow [0, 0]$$

$$\frac{\partial}{\partial x}(\dots) = 0, \quad \text{a.k.a. flow is the same everywhere along the plate}$$

Looking at the x-momentum equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (1.1)$$

We see immediately there are many terms that drop out due to the constant nature along the plate leaving us with

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \quad (1.2)$$

1.2 Part B

Here we define $\eta = By/t^n$, $u = t^m Af(\eta)$, and $t^m Cg(\eta)$. We can define derivatives as follows

$$\frac{\partial u}{\partial t} = -nmt^{m-1}f' \frac{AB}{t^{n+1}} \quad (1.3)$$

$$\frac{\partial u}{\partial y} = t^m A f' \frac{B}{t^n} \quad (1.4)$$

$$\frac{\partial^2 u}{\partial y^2} = t^m A f'' \frac{B}{t^n} \quad (1.5)$$

We can then plug this in to the x-momentum equation in an attempt to find a similarity solution

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \quad (1.6)$$

Plugging this in we find that it reduces to

$$-nmt^{m-n-2}f'ABy + t^{2m-n}Cgf'B = \nu t^{m-n}Af''B \quad (1.7)$$

From this we see that there is no way to cancel out the time dependence in the equation meaning that there is no similarity solution for this particular configuration.

Problem 2

In this question we are asked to use a shooting method to numerically solve the Blasius Equation.

$$f''' + \frac{1}{2}ff'' = 0 \quad (2.1)$$

$$\text{B.C.s } f'(0) = 0, \quad f(0) = 0, \quad f'(\eta) \rightarrow 0 \text{ as } \eta \rightarrow \infty$$

To do this we can use numeric approximations with a forward differencing scheme.

$$f'_i = \frac{f_{i+1} - f_i}{\Delta\eta} \Rightarrow f_{i+1} = f_i + \Delta\eta f'_i \quad (2.2)$$

We can do similar things for f'' and f''' , and use eq. (2.1) to form the equation to solve for each subsequent f''

$$f''_{i+1} = f''_i + \Delta\eta \left(-\frac{1}{2}ff'' \right) \quad (2.3)$$

Using the provided initial guess of $f''(0) = 0.2$ yields a result of $f'(10) = 0.72479...$, and given the boundary conditions given we know this is not correct. Massaging the initial guess to a value of $f''(0) = 0.3155$ gives an $f'(10) = 1.00624...$, which is significantly closer. Next we adjust the $\Delta\eta$ in the method going from 0.1 to 0.01. This is because we are given that the shear is known to be $f''(0) = 0.332$, and reducing the step size allows us to get closer to this value. Upon changing the $\Delta\eta$ and changing the initial guess to $f''(0) = 0.3305$ leads to the freestream condition of $f'(10) = 1.00389...$

Plots of all three parts are shown in fig. 2.1 and all three show generally the same behavior, but the second and third configurations reflect the B.C.s defined.

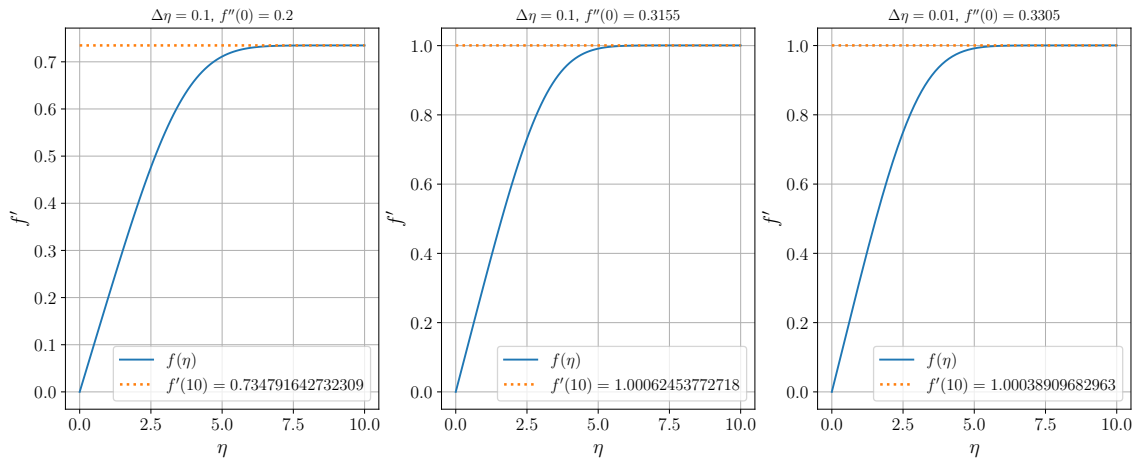


Figure 2.1: Shooting Method with Three Different Configurations

Problem 3

3.1 Part A

3.1.1 Parabolic

We begin by deriving the parabolic approximation of the velocity profile.

$$\frac{u(y)}{U_\infty} = a_0 + a_1\eta + a_2\eta^2 \quad \eta = \frac{y}{\delta} \quad (3.1)$$

$$\text{B.C.s } u(0) = 0, \quad u(\delta) = U_\infty, \quad \left. \frac{\partial u}{\partial y} \right|_\delta$$

Applying these boundary conditions shows that $a_0 = 0$. The rest of the boundary conditions set up the system of equations as shown.

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (3.2)$$

This can be evaluated to find

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad (3.3)$$

Thus the parabolic approximation takes the form

$$\frac{u(y)}{U_\infty} = 2\eta - \eta^2 \quad (3.4)$$

3.1.2 $\mathcal{O}(\eta^3)$ with Wall Boundary condition

A very similar process can be done for each $\mathcal{O}(\eta^3)$ function, but we will impose different boundary conditions.

$$\frac{u(y)}{U_\infty} = a_0 + a_1\eta + a_2\eta^2 + a_3\eta^3 \quad (3.5)$$

$$\text{B.C.s } u(0) = 0, \quad u(\delta) = U_\infty, \quad \left. \frac{\partial u}{\partial y} \right|_\delta, \quad \left. \frac{\partial^2 u}{\partial y^2} \right|_0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (3.6)$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix} \quad (3.7)$$

$$\frac{u(y)}{U_\infty} = \frac{3}{2}\eta - \frac{1}{2}\eta^3 \quad (3.8)$$

3.1.3 $\mathcal{O}(\eta^3)$ with Freestream Boundary Condition

Finally we impose the final boundary condition where there is zero curvature at $y = \delta$

$$\frac{u(y)}{U_\infty} = a_0 + a_1\eta + a_2\eta^2 + a_3\eta^3 \quad (3.9)$$

$$\text{B.C.s } u(0) = 0, \quad u(\delta) = U_\infty, \quad \left. \frac{\partial u}{\partial y} \right|_\delta, \quad \left. \frac{\partial^2 u}{\partial y^2} \right|_\delta$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 2 & 6 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (3.10)$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 1 \end{bmatrix} \quad (3.11)$$

$$\frac{u(y)}{U_\infty} = 3\eta - 3\eta^2 + \eta^3 \quad (3.12)$$

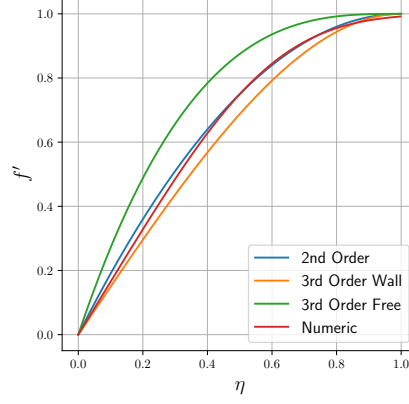


Figure 3.1: Various Velocity Profiles for Flow Over a Flat Plate

3.2 Part B

Graphing all the results along with our numeric results from Problem 2 yields the results seen in fig. 3.1

3.3 Part C

The momentum integral equation is as follows

$$\frac{U_\infty \theta}{\nu} \left[\frac{d\theta}{dx} + \frac{2\theta + \delta^*}{U_\infty} - \frac{V_w}{U_\infty} = \frac{\tau_w}{\rho U_\infty^2} \right] \quad (3.13)$$

We can define new variables H , S , and λ that can be used to scale the momentum integral equation

$$H = \frac{\delta^*}{\theta} \quad \lambda = \frac{\theta^2}{\nu} \frac{dU_\infty}{dx} \quad S = \frac{\tau_w \theta}{\mu U_\infty}$$

This transforms the MIE to

$$\frac{U_\infty}{2} \frac{d}{dx} \left(\frac{\theta^2}{\nu} \right) + (H + 2)\lambda = S \quad (3.14)$$

Following Thwaites' Method to find the momentum thickness as a function of x

$$\theta^2(x) = \frac{0.45\nu}{U_\infty^6} \int_0^x U_\infty^5 dx \quad (3.15)$$

For the two third order profiles we can directly compute δ^* and θ using their respective equations

$$\delta^* = \int_0^\infty \left(1 - \frac{u(y)}{U_\infty}\right) dy \quad (3.16)$$

$$\rho U_\infty^2 \theta = \rho \int_0^\infty u(y)(U_\infty - u(y)) dy \quad (3.17)$$

Because $u(\delta) = U_\infty$ the integrals go to 0 from δ to ∞ and thus we can truncate the integral at δ . For the first 3rd order approximation this gives us the integrals

$$\delta^* = \int_0^\delta \left(1 - \frac{3}{2}\eta + \frac{1}{2}\eta^3\right) dy \quad (3.18)$$

$$\theta = \frac{1}{U_\infty} \int_0^\delta \left(\frac{3}{2}\eta - \frac{1}{2}\eta^3\right) \left(1 - \frac{3}{2}\eta + \frac{1}{2}\eta^3\right) dy \quad (3.19)$$

Using Wolfram Alpha to solve we get that

$$\delta^* = \frac{3}{8}\delta \text{ and } \theta = \frac{39}{280} \frac{\delta}{U_\infty} \quad (3.20)$$

for the 3rd order approximation with the bounding condition at the wall

Returning to the MIE we see that there is a derivative term of the freestream velocity which is constant. This then means that this term goes to zero and we are left with

$$\frac{d\theta}{dx} = \frac{C_f}{2} \quad (3.21)$$

We also know that

$$\frac{C_f}{2} = \frac{\tau_w}{\rho U_\infty^2} \quad (3.22)$$

Shear stress is defined as

$$\tau_w = \mu \frac{\partial u}{\partial y} \Big|_{y=0} = \mu \left(\frac{3}{2} \frac{1}{\delta} - \frac{3}{2} \frac{y^2}{\delta^3} \right) \Big|_{y=0} = \mu \frac{3}{2} \frac{1}{\delta} \quad (3.23)$$

Thus θ can be computed as follows

$$\frac{d\theta}{dx} = \frac{1}{\rho U_\infty^2} \left(\frac{3\mu}{2\delta} \right) \quad (3.24)$$

$$\theta(x) = \frac{3\nu}{2\delta U_\infty^2} x \quad (3.25)$$

The second equation with the bounding condition at the edge of the boundary layer is solved similarly with the integrals

$$\int_0^\delta (1 - (3\eta - 3\eta^2 + \eta^3)) dy \quad (3.26)$$

$$\theta = \frac{1}{U_\infty} \int_0^\delta (3\eta - 3\eta^2 + \eta^3) (1 - (3\eta - 3\eta^2 + \eta^3)) dy \quad (3.27)$$

which can once again be solved to find that

$$\delta^* = \frac{\delta}{4} \quad \theta = \frac{3}{28} \delta \quad (3.28)$$

The shear stress for this approximation can be solved as follows

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \mu \left(3\frac{1}{\delta} - 6\frac{y}{\delta^2} + 3\frac{y^2}{\delta^3} \right) \Big|_{y=0} = 3\mu \frac{1}{\delta} \quad (3.29)$$

3.4 Part D

Both third order profiles deviate from the numeric solution more than the parabolic solution. The third order scheme with the wall boundary condition trends towards a slightly more adverse pressure gradient than the numeric solution. The other third order scheme deviates more with a pressure gradient that is more favorable than reality.